statshw4

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Homework 4

conceptual exercises 2,3,9 Hand in 19, 26, 27, 28, 29, 31, 32

Question 19

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(Sleuth3)

#Part a
t.test(Humerus ~ Status, data=ex0221)
```

```
##
##  Welch Two Sample t-test
##
## data:  Humerus by Status
## t = -1.7207, df = 43.824, p-value = 0.09236
## alternative hypothesis: true difference in means between group Perished and group Survived is not equ
## 95 percent confidence interval:
##  -0.021894675  0.001728008
## sample estimates:
## mean in group Perished mean in group Survived
##              0.7279167              0.7380000
```

```r
#Part b
#t.tests are based on a normal data spread assumption.

#Part c
#Yes, a continunity correction would be needed for the Z-statistic.

#Part d
wilcox.test(Humerus ~ Status, data=ex0221)
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  Humerus by Status
## W = 331, p-value = 0.1718
## alternative hypothesis: true location shift is not equal to 0
```

```r
min(ex0221$Humerus)
```

```
## [1] 0.659
```

```r
birds_wo_smallest <- ex0221 |> filter(Humerus > 0.659)

t.test(Humerus ~ Status, data=birds_wo_smallest)
```

```
##
##  Welch Two Sample t-test
##
## data:  Humerus by Status
## t = -1.373, df = 48.967, p-value = 0.176
## alternative hypothesis: true difference in means between group Perished and group Survived is not equ
## 95 percent confidence interval:
##  -0.017460170  0.003286257
## sample estimates:
## mean in group Perished mean in group Survived
##               0.730913               0.738000
```

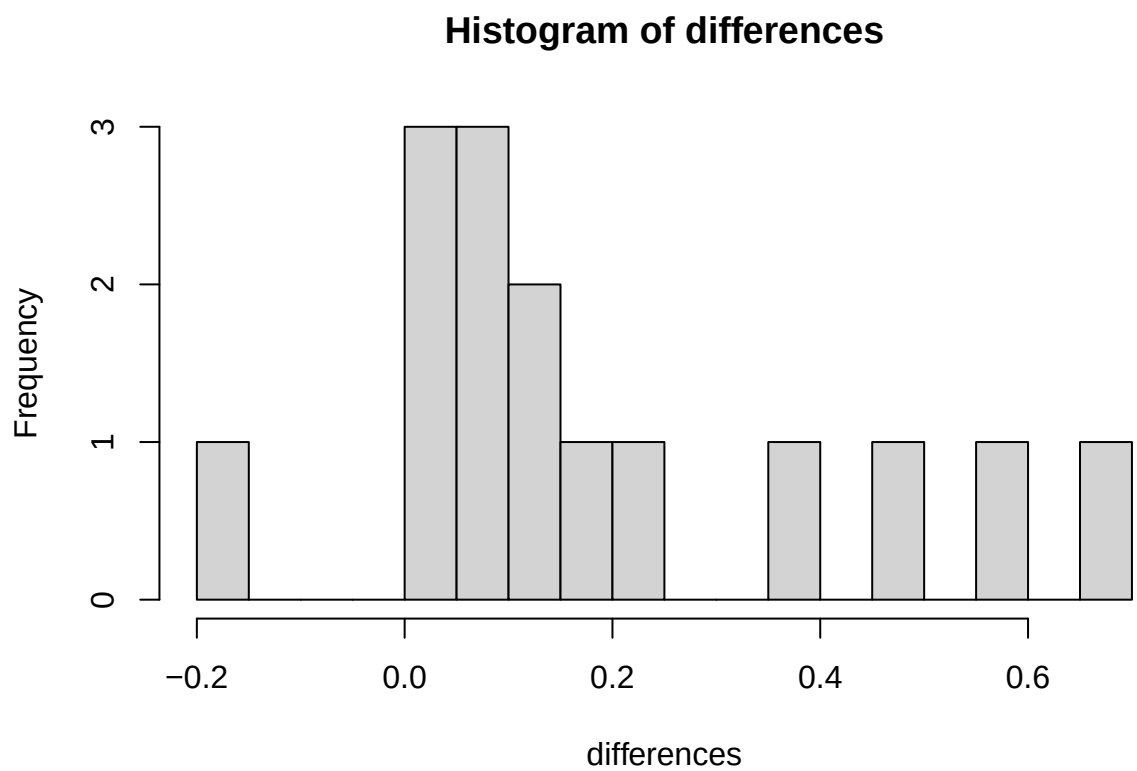
```r
wilcox.test(Humerus ~ Status, data=birds_wo_smallest)
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

##
##  Wilcoxon rank sum test with continuity correction
##
## data:  Humerus by Status
## W = 331, p-value = 0.2588
## alternative hypothesis: true location shift is not equal to 0
```

```r
#At first, the t.test pvalue = 0.09236 and the wilcox.test = 0.1718
#After removing the smallest value, the t.test pvalue =  0.176 and
#the wilcox.test = 0.2588

#part e
#Wilcox tests are resiliant to outliers, while t.test are not, shown by
#removing the outlier and the t.test pvalue almost doubling.
```

Question 26

```r
#Part a
differences <- case0202$Unaffected - case0202$Affected
hist(differences, breaks = 20)
```

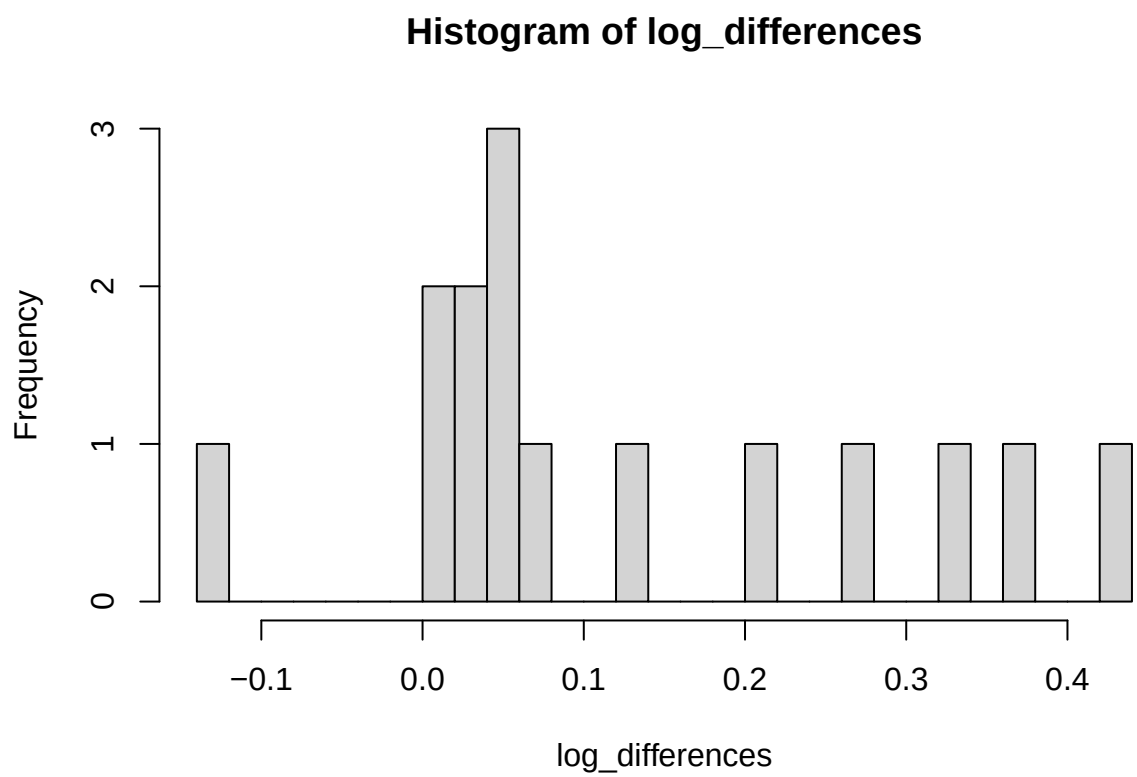



```r
t.test(case0202$Unaffected, case0202$Affected, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  case0202$Unaffected and case0202$Affected
## t = 3.2289, df = 14, p-value = 0.006062
## alternative hypothesis: true mean difference is not equal to 0
```

```
## 95 percent confidence interval:
##  0.0667041 0.3306292
## sample estimates:
## mean difference
##       0.1986667
```

```r
# There appears to be a right skew of the differences.

#Part B
# Compute the log-transformed hippocampus volumes
log_affected <- log(case0202$Affected)
log_unaffected <- log(case0202$Unaffected)

log_differences <- log_unaffected - log_affected
hist(log_differences, breaks = 20)
```




```r
#The data spread does not appear to be more symmetric

#Part c
t.test(log_unaffected, log_affected, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  log_unaffected and log_affected
```

```
## t = 3.1967, df = 14, p-value = 0.006463
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.04228227 0.21470597
## sample estimates:
## mean difference
##       0.1284941
```

```r
#The pvalue after a log transformation is 0.006463, compared to a 0.006062
#before transforming the data, a very small difference.

#Part d
mean_log_diff <- mean(log_differences)
se_log_diff <- sd(log_differences) / sqrt(length(log_differences))

t_critical <- qt(0.975, df = length(log_differences) - 1)
ci_lower_log <- mean_log_diff - t_critical * se_log_diff
ci_upper_log <- mean_log_diff + t_critical * se_log_diff

median_ratio_estimate <- exp(mean_log_diff)
ci_lower_ratio <- exp(ci_lower_log)
ci_upper_ratio <- exp(ci_upper_log)

cat("Mean Log Difference:", mean_log_diff, "\n")
```

```
## Mean Log Difference: 0.1284941
```

```r
cat("95% CI for Log Difference: (", ci_lower_log, ",", ci_upper_log, ")\n")
```

```
## 95% CI for Log Difference: ( 0.04228227 , 0.214706 )
```

```r
cat("Estimated Median Ratio:", median_ratio_estimate, "\n")
```

```
## Estimated Median Ratio: 1.137115
```

```r
cat("95% CI for Median Ratio: (", ci_lower_ratio, ",", ci_upper_ratio, ")\n")
```

```
## 95% CI for Median Ratio: ( 1.043189 , 1.239497 )
```

Question 27

```r
wilcox.test(log_affected, log_unaffected, paired = TRUE)
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  log_affected and log_unaffected
## V = 9, p-value = 0.002014
## alternative hypothesis: true location shift is not equal to 0
```

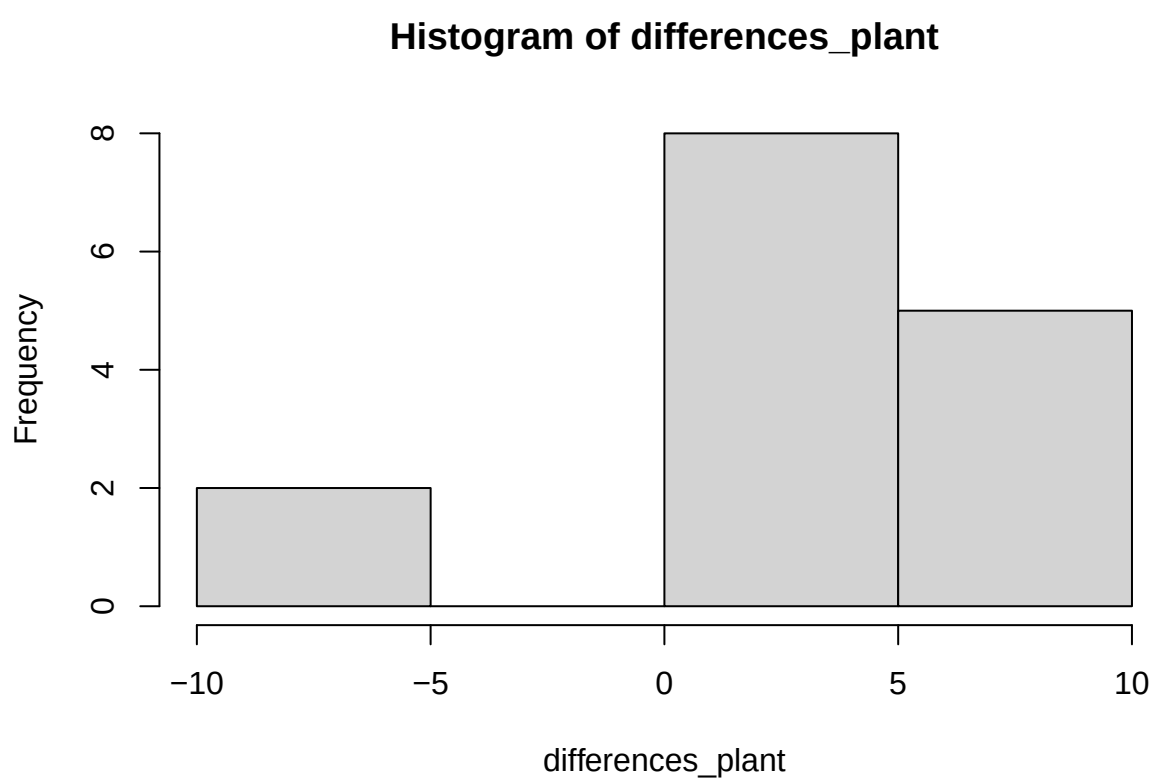
```r
wilcox.test(case0202$Affected, case0202$Unaffected, paired = TRUE)
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  case0202$Affected and case0202$Unaffected
## V = 9, p-value = 0.002014
## alternative hypothesis: true location shift is not equal to 0
```

```r
#The pvalue of the two tests above match because wilcox.test doesn't change with
# a log transformation, because the ranking is the same.

#It doesn't seem apparent that the assumptions of the rank test are any more
#appropiate on these scales
```

Question 28

```r
#Part a
differences_plant <- ex0428$Cross - ex0428$Self
hist(differences_plant)
```



```r
#Part b
t.test(ex0428$Self, ex0428$Cross, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  ex0428$Self and ex0428$Cross
## t = -2.1474, df = 14, p-value = 0.04976
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -5.230168289 -0.003165044
## sample estimates:
## mean difference
##       -2.616667
```

```r
#Part c
t.test(ex0428$Cross, ex0428$Self, paired = TRUE, conf.level = 0.95)
```

```
##
##  Paired t-test
##
## data:  ex0428$Cross and ex0428$Self
## t = 2.1474, df = 14, p-value = 0.04976
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.003165044 5.230168289
## sample estimates:
## mean difference
##        2.616667
```

```r
#I believe this returns the correct confidence interval, I did a different
#method for an earlier question.

#Part D
#The histogram of part a is suggesting quite a large discrepancy between
#cross and self fertilized plants, while the pvalues of our tests are not or
#barely reaching significance.

#Part e
wilcox.test(ex0428$Self, ex0428$Cross, paired = TRUE)
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  ex0428$Self and ex0428$Cross
## V = 24, p-value = 0.04126
## alternative hypothesis: true location shift is not equal to 0
```

Question 29

```r
#Part a
salvage <- ex0429
ex0429$Action <- as.integer(ex0429$Action)
wilcox.test(ex0429$Action, ex0429$PercentLost)
```

```
## Warning in wilcox.test.default(ex0429$Action, ex0429$PercentLost): cannot
## compute exact p-value with ties
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  ex0429$Action and ex0429$PercentLost
## W = 32, p-value = 0.0002534
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(ex0429$Action, ex0429$PercentLost, conf.int = TRUE, conf.level = .95)
```

```
## Warning in wilcox.test.default(ex0429$Action, ex0429$PercentLost, conf.int =
## TRUE, : cannot compute exact p-value with ties
```

```
## Warning in wilcox.test.default(ex0429$Action, ex0429$PercentLost, conf.int =
## TRUE, : cannot compute exact confidence intervals with ties
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  ex0429$Action and ex0429$PercentLost
## W = 32, p-value = 0.0002534
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
##  -51.10005 -17.20003
## sample estimates:
## difference in location
##              -40.51043
```

```r
#Part b
t.test(PercentLost ~ Action, data = ex0429, conf.level = 0.95)
```

```
##
##  Welch Two Sample t-test
##
## data:  PercentLost by Action
## t = 3.1091, df = 12.236, p-value = 0.008849
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  11.45841 64.75111
## sample estimates:
## mean in group 1 mean in group 2
##        54.83333        16.72857
```

Between part a and part b, the pvalues were 0.0002534 and 0.008849 respectively. This indicates that the t.test might be missing a problem with the data, possibly an outlier that is skewing significance. Both conclude a strong confidence level.

##Question 30

```
sun <- ex0430
#because I do not see any obvious outliers, I will be using a t.test analysis.
sunscreen <- ex0430$Sunscreen
before <- ex0430$PreTreatment

t.test(before, sunscreen, paired = TRUE, conf.level = .95)
```

```
##
##  Paired t-test
##
## data:  before and sunscreen
## t = -10.015, df = 12, p-value = 3.525e-07
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -302.0497 -194.1042
## sample estimates:
## mean difference
##       -248.0769
```

Personally, there are so many things I will consider when I am tolerating the sun that aren't sunscreen, such as hydration, clothing coverage, direct or indirect sun exposure, outside/ inside, how humid it is, how mch time between the first test, applying, the sunscreen, and going back into the sun, etc. The genetic makeup/ pre-existing conditions could have a big impact too, as someone with a heart condition might not stay in the sun as long the second round. I'm fairly certain women have higher pain tolerances, but worse body temperature regulation.